

# INTERNATIONAL STANDARD



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## Event video data recorder for road vehicle accidents – Part 1: Basic requirements



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## Event video data recorder for road vehicle accidents – Part 1: Basic requirements

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The text of this International Standard is based on the following documents:

| CDV          | Report on voting |
|--------------|------------------|
| 100/2839/CDV | 100/2947/RVC     |

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

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## INTRODUCTION

There is a distinction between event video data recorder (EVDR) systems and digital tachographs (DTG, ISO 16844-7), which record vehicle dynamics and the driver's activities during the entire driving period. There is also a distinction between EVDR systems and event data recorders (EDR, IEEE 1616), which record vehicle dynamics and the driver's activities before, during and after the event. DTGs and EDRs both have direct connections to the vehicle's internal signal line, such as in-vehicle network (IVN) or analogue signal line, whereas direct connection is not required for EVDRs.

EVDR systems are prohibited from serving purposes other than the aforementioned. They cannot be placed in unapproved areas and/or not record sound.



# EVENT VIDEO DATA RECORDER FOR ROAD VEHICLE ACCIDENTS –

## Part 1: Basic requirements

### 1 Scope

This part of IEC 63005 describes basic requirements for event video data recorders (EVDRs) for road vehicle accidents, used for identifying and analysing causes of accidents based on video from a front-mounted camera and other information obtained before and after such events. In addition to video from a front-mounted camera and vehicle behaviour, these products can record side and/or rear video data for enhanced functionalities in determining causes of accidents and analysing collision events.

### 2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60068-2-1:2007, *Environmental testing – Part 2-1: Tests – Test A: Cold*

IEC 60068-2-6, *Environmental testing – Part 2-6: Tests – Test Fc: Vibration (sinusoidal)*

IEC 60068-2-27:2010, *Environmental testing – Part 2-27: Tests – Test Ea and guidance: Shock*

ISO 12233, *Photography – Electronic still picture imaging — Resolution and spatial frequency responses*

### 3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

#### 3.1

##### **EVDR for road vehicle accidents**

system that stores vehicle video data of the accident on an electronic recording medium before, during, and after collision accident events with other vehicles, with passers-by and with any other objects

#### 3.2

##### **event data**

information recorded by the EVDR to facilitate analysis of accident scenarios in the case of collision accident events with other vehicles, pedestrians or objects

Note 1 to entry: The term refers to all videos and additional information before, during, and after collision.

### **3.3**

#### **video input information**

video data before, during, and after collision that contains video data from the front camera

### **3.4**

#### **vehicle dynamics data**

information on a vehicle's dynamic behaviour such as acceleration, angular velocity, and physical quantities related to collision

### **3.5**

#### **integrity verification value**

information used to detect doctoring and/or deletion of event data

### **3.6**

#### **DTG for road vehicles**

#### **digital tachograph for road vehicles**

device that adheres to guidelines on driving records and devices defined in ISO 16844-7

### **3.7**

#### **EDR for road vehicles**

#### **event data recorder for road vehicles**

system that adheres to IEEE 1616

### **3.8**

#### **interlinked video data recording system for road vehicle accidents**

EVDR system interlinked to the DTG or EDR

### **3.9**

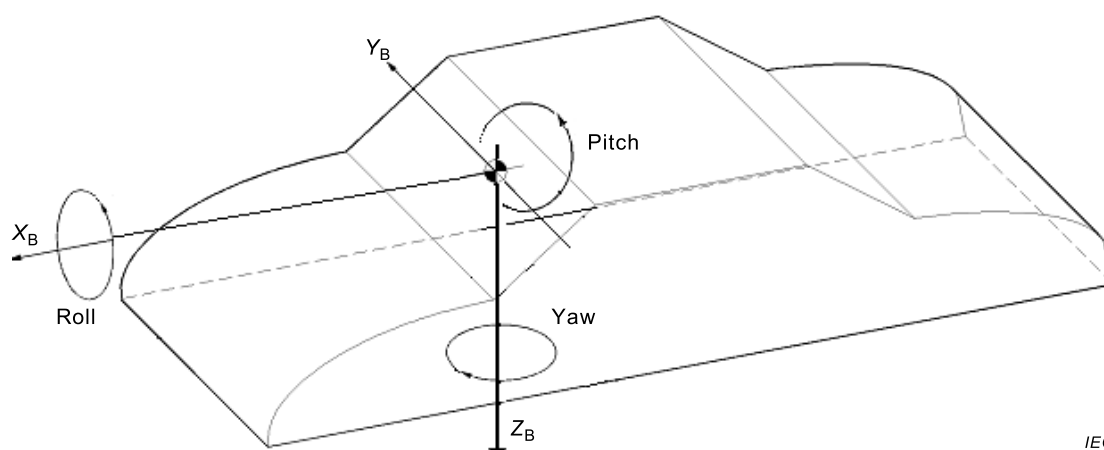
#### **independent EVDR for road vehicle accidents**

EVDR that operates independently from a DTG or EDR

## **4 Abbreviated terms and symbols**

For vehicle orientation related symbols, see Figure 1.

|        |                                                                                |
|--------|--------------------------------------------------------------------------------|
| DTG    | digital tachograph                                                             |
| EDR    | event data recorder                                                            |
| EVDR   | event video data recorder                                                      |
| FOV    | field of view                                                                  |
| fps    | frames per second                                                              |
| g      | gravitational acceleration                                                     |
| GPS    | Global Positioning System                                                      |
| GNSS   | global navigation satellite system                                             |
| IVN    | in-vehicle network                                                             |
| SFR    | spacial frequency response                                                     |
| $a_x$  | acceleration in the $X_B$ direction (front and rear) in body fixed coordinates |
| $a_y$  | acceleration in the $Y_B$ direction (left and right) in body fixed coordinates |
| $a_z$  | acceleration in the $Z_B$ direction (vertical) in body fixed coordinates       |
| $\psi$ | angular velocity around the $Z_B$ axis in body fixed coordinates (yaw rate)    |



**Figure 1 – Standard coordinate system of a vehicle equipped with the EVDR (body fixed coordinates)**

## 5 Types of EVDR

### 5.1 Classification by security level

EVDRs can be classified into two types listed below depending on the security level.

- 1) General type: EVDR without video-data integrity-checking function for stored event data.
- 2) Enhanced security type: EVDR equipped with video-data integrity verification function for stored event data.

### 5.2 Classification by interoperability of functions

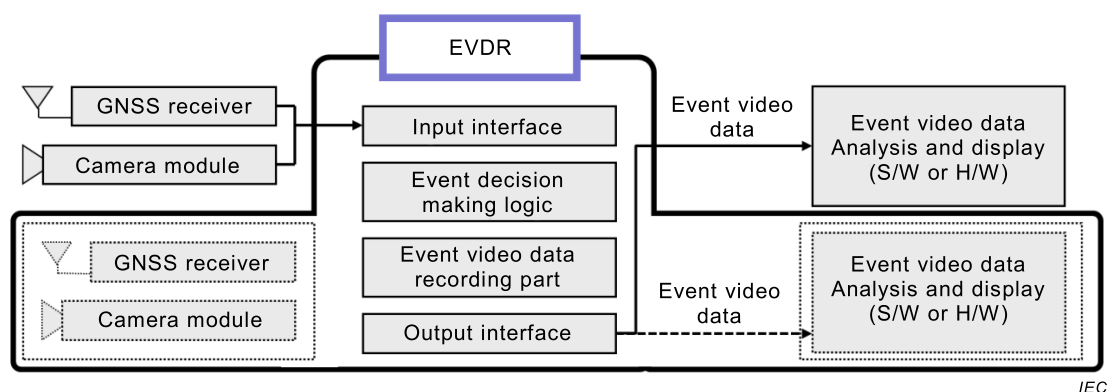
EVDRs can be classified into two types listed below depending on interoperability with DTG or EDR.

- 1) Independent: EVDRs that operate independently from DTGs or EDRs.
- 2) Interlinked: EVDRs that operate interlinked to DTGs or EDRs.

## 6 Functional requirements

### 6.1 Basic functions of the EVDR for road vehicle accidents

The primary purpose of EVDRs is to record video data before, during and after the event. For EVDR systems, the direct connection to the vehicle's internal signal line such as the in-vehicle network (IVN) or the analogue signal line is not required. Only the power line of the EVDR is connected to the power line of the installed vehicle.



**Figure 2 – Basic functions of the EVDR for road vehicle accidents**

The EVDR shall have the following basic functions.

- a) Input interface function that accepts input from the video acquisition device and the GNSS receiver; the GNSS receiver and/or video acquisition device can be either included in or separated from the EVDR main system (see Figure 2).
- b) Decision-making function to check the accident event: the EVDR checks if an accident event has occurred based on the vector sum of acceleration in the  $X_A$  and  $X_B$  directions, expressed as  $a_x$  and  $a_y$ , or based on the difference of acceleration per unit of time. For interlinked EVDR systems, the decision-making function to check the accident event can be located in the interlinked DTG or EDR.
- c) Recording and maintaining functions for event video data:
  - 1) general type: safely records and maintains event video data;
  - 2) enhanced security type: safely records and maintains event video data while preventing doctoring and deletion of the event data.
- d) Output interface function to transmit stored event video data: the DTG or EDR of the interlinked EVDR system can have the following functions. Since the interoperability function is enabled, the same functions are not required to be included in the EVDR:
  - 1) accident event decision making function;
  - 2) event data recording function;
  - 3) event data storing function;
  - 4) sensor function including GNSS for acquisition of event data.

### 6.2 Types of event data

#### 6.2.1 General

The types of event data to be recorded by the EVDR are as shown in Table 1.

**Table 1 – Types of event data**

| Types of event data to be recorded | Mandatory event data                                                                   | Optional event data                    |
|------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------|
| Video input information            | Front camera video (Including video generation time and product identification number) | Side camera video<br>Rear camera video |
| Vehicle behaviour information      | Acceleration $a_x$<br>Acceleration $a_y$                                               | Yaw angular velocity $\psi$            |
| Other information                  | Event occurrence time                                                                  | GNSS information                       |

### 6.2.2 Mandatory event data

The types of event data to be recorded by the EVDR can be classified into "mandatory event data" and "optional event data." It is essential for EVDRs to record the following "mandatory event data" listed in Table 2.

**Table 2 – Mandatory event data items**

| Event data item      |                               |                | Resolution                                                  | Data measurement range | Data storing term |             | Minimum sampling rate |
|----------------------|-------------------------------|----------------|-------------------------------------------------------------|------------------------|-------------------|-------------|-----------------------|
|                      |                               |                |                                                             |                        | Before event      | After event |                       |
| Vehicle dynamic data | Acceleration $a_x$            |                | Resolving power<br>< 0,01 g                                 | < -2 g<br>> 2 g        | 10 s              | 10 s        | 20 Hz                 |
|                      | Acceleration $a_y$            |                | Resolving power<br>< 0,01 g                                 | < -2 g<br>> 2 g        | 10 s              | 10 s        | 20 Hz                 |
| Video data           | Front camera video            | Resolution     | > 900 K pixels                                              | -                      | 10 s              | 10 s        | 20 fps                |
|                      |                               | Horizontal FOV | -                                                           | > 80° <sup>a</sup>     |                   |             |                       |
|                      |                               | Vertical FOV   | -                                                           | > 50° <sup>a</sup>     |                   |             |                       |
|                      |                               | Colour/<br>B&W | -                                                           | Colour                 |                   |             |                       |
| Other data           | Event occurrence time         |                | ≤ 1 s<br>(including year, month, day, time, minute, second) | -                      | -                 | -           | Event-based           |
|                      | Product identification number |                | -                                                           | -                      | -                 | -           | Event-based           |

<sup>a</sup> Specifications provided by respective manufacturers take precedence if the above conditions are met. If external power supply to the EVDR is interrupted due to shock or other unforeseen circumstances, storing time conditions need not be fulfilled.

Mandatory items to be stored shall have the following specifications:

- Acceleration  $a_x$ : this refers to the acceleration in the longitudinal direction (X direction in the body-fixed coordinate system) of the vehicle. The data storage term shall include 10 s before the event and 10 s after the event. The sampling rate shall be greater than 20 Hz, and the resolution shall be finer than 0,01 g.

- Acceleration  $a_y$ : this refers to the acceleration in the transverse direction (Y direction in the body-fixed coordinate system) of the vehicle. The data storage term shall include 10 s before the event and 10 s after the event. The sampling rate shall be greater than 20 Hz, and the resolution shall be finer than 0,01 g.
- Front camera video: this refers to video data recorded by the front camera. The video data shall be recorded at a resolution greater than 900 000 pixels, the horizontal FOV shall be greater than 80°, the vertical FOV shall be greater than 50°, and the frame rate shall be greater than 20 fps. The data storage term shall include 10 s before the event and 10 s after the event. To maintain the desired frame rate, the sampling rate shall be greater than 20 Hz. However, frames that remain unchanged are not included when counting the sampling rate.
- Event occurrence time: this is the time at which the event is detected by the accelerometer(s). The event data for a minimum of three occurrences of events shall be recorded. Event data to be recorded shall include details on year, month, day, time, minutes, and seconds, and have a minimum period of 20 Hz. To ensure accurate recording of event occurrence time, the internal clock of the EVDR shall operate continuously regardless of the state of the vehicle's ignition.
- Product identification number: these numbers shall be assigned as specified in Annex B.

### 6.2.3 Optional event data

Items listed under "optional event data" are given in Table 3.

**Table 3 – Optional event data items**

| Event data item       |                             | Resolution       | Data measurement range | Data storing time |             | Minimum sampling rate |
|-----------------------|-----------------------------|------------------|------------------------|-------------------|-------------|-----------------------|
|                       |                             |                  |                        | Before event      | After event |                       |
| Vehicle dynamics data | Yaw angular velocity $\psi$ | 0,2 °/s          | –180 °/s ~ 180 °/s     | 10 s              | 10 s        | 20 Hz                 |
| Video input data      | Side camera video           | > 300 000 pixels | -                      | 10 s              | 10 s        | 10 fps                |
|                       | Rear camera video           |                  |                        |                   |             |                       |
| Other data            | GNSS data                   | -                | -                      | 10 s              | 10 s        | 1 Hz                  |

Items to be stored optionally shall have the following specifications.

- Yaw angular velocity  $\psi$ : this represents the angular velocity around the vertical Z-axis (Z direction in body-fixed coordinate system) of the vehicle. The data storage term shall include 10 s before the event and 10 s after the event. The sampling rate shall be greater than 20 Hz. Data measurement range shall be from –180 °/s to 180 °/s or more, and the resolution shall be finer than 0,2 °/s.
- Side camera video: this refers to video data recorded by the side camera. The resolution of the recorded data shall be greater than 300 000 pixels. The data storage term shall include 10 s before the event and 10 s after the event. The sampling rate shall be greater than 10 fps.
- Rear camera video: this refers to video data recorded by the rear camera. The resolution of the recorded data shall be greater than 300 000 pixels. The data storage term shall include 10 s before the event and 10 s after the event. The sampling rate shall be greater than 10 fps.
- GNSS data: this is the latitude, longitude and time information of the vehicle as measured by the GNSS. The data storage term shall include 10 s before the event and 10 s after the event. The sampling rate shall be greater than one session per second.

## 6.3 Power supply unit and electrical safety

### 6.3.1 Main power

The main supply power to operate the EVDR shall be the vehicle's electrical power connected to the battery; power voltage details are given in Table 4. For limited-time operation, the internal battery of the EDVR can be used.

**Table 4 – Operating voltage of EVDR under test**

| Operating voltage<br>V | Test voltage<br>V |
|------------------------|-------------------|
| 12                     | 13,5 ± 0,5        |
| 24                     | 27,0 ± 1,0        |

### 6.3.2 Fuse

If fuses are used for the power supply line of the EVDR, rated current values shall be specified in the manual.

### 6.3.3 Protection from power failure

In case of electrical power failure, the EVDR shall protect all previously stored data.

## 7 General requirements

### 7.1 Performance for storing acceleration

The vector sum of  $a_x$  and  $a_y$  stored in the EVDR shall be able to have a maximum value within the range of the measurement data range for the compatibility of event data.

### 7.2 Physical structure

#### 7.2.1 General

When installed, the EVDR shall not block the front view of the driver.

The EVDR shall be mounted so as not to take video data of passengers. The EVDR shall be installed with the front facing the forward direction so that the video and audio data of the cabin will not be recorded.

The EVDR shall provide audible or visual alerts to the user if the event data cannot be stored due to a malfunction or to the limitation of storage capacity.

Event data shall not be lost even in cases of power failure during collision events, and at least the data from the three most recent events shall be preserved.

All mandatory and optional storage items shall be stored with clear indication of event time.

Event data can be stored in an external or internal storage medium; enhanced security type EVDRs shall satisfy the requirements of 7.4.

If event data is stored in interlinked DTG or EDR as required by this document, the interlinked EVDR is considered to meet the storage requirements.

Event data and front camera information shall include the event occurrence time and product identification number that is assigned under the guidelines of Annex B.

### **7.2.2 Input interface**

The input interface consists of the camera module and interface with the GNSS receiver; the detail specifications are dependent upon the manufacturers of the EVDRs.

## **7.3 Video camera performance**

### **7.3.1 Basic camera performance**

Video data stored in the EVDR shall be downloaded onto a PC as outlined in the manual. The video data stored in the EVDR can also be viewed and analysed on the EVDR itself. The video data shall be checked if it meets the requirements listed in Table 2 for resolution, colour and sampling rate.

The event video data shall include product identification number and event occurrence time.

### **7.3.2 Camera FOV**

EVDRs shall have the horizontal and vertical FOV specified in 6.2.2. The front camera video data shall have a minimum horizontal FOV of 80°, a minimum vertical FOV of 50°. Resolution shall be greater than 900 000 pixels. The front camera video shall include video data of all four corners of the square described in the test of Annex A with a horizontal angle of 80° and a vertical angle of 50°. Details on the FOV test equipment are given in Annex A.

### **7.3.3 Vehicle registration plate detection performance**

The camera of the EVDR shall be capable of detecting the registration plates of vehicles. The user converts the stored video data into still images, and performs spatial frequency response (SFR) analysis to ensure that the resolution is greater than 278 lines. The resolution test chart and SFR calculations shall conform to the guidelines of ISO 12233 given in Annex C.

## **7.4 Data storage and security**

Integrity verification values shall be generated for event data, and doctoring/deletion detection functions shall be provided.

## **7.5 Environmental reliability tests**

### **7.5.1 General**

For environmental reliability tests, the mandatory storage items listed in Table 2 shall be fulfilled.

### **7.5.2 Low-temperature operation test**

After operating the video data recorder at a temperature of  $(-20 \pm 3) ^\circ\text{C}$  with the power switched off for 2 h and on for 30 min over a period of 16 h, and placing that recorder at room temperature for one hour, the video data recorder shall be corrosion-free and capable of normal operation, and among the mandatory items to be stored listed in Table 2, video data information shall be satisfied. Note that the test specification below refers to IEC 60068-2-1.

### **7.5.3 Operation at high-temperature**

After operating the EVDR at a temperature of  $(70 \pm 2) ^\circ\text{C}$  with the power switched off for 2 h and on for 30 min over a period of 16 h, place the EVDR at room temperature for 1 h. The EVDR shall be corrosion-free and capable of normal operation. The mandatory items to be



stored listed in Table 2 shall be satisfied. Note that the test specification below refers to IEC 60068-2-1.

#### 7.5.4 High temperature storage test

After placing the EVDR at the temperature of  $(85 \pm 2) ^\circ\text{C}$  over a period of 16 h, place the EVDR at room temperature for 1 h. The EVDR shall be corrosion-free and capable of normal operation and among mandatory items to be stored listed in Table 2, video data information shall be satisfied. Note that the test specification below refers to IEC 60068-2-1.

#### 7.5.5 Vibration test

The EVDR shall pass the vibration test. Note that the vibration test specification in Table 5 refers to IEC 60068-2-6.

- The vibration applied to the EVDR shall be applied in X, Y and Z directions separately. The applied vibration condition is listed in Table 5.
- After dismounting the EVDR from the vibration test equipment, the EVDR shall not have any separation of components. This visual inspection is done by the test operator.
- The EVDR shall satisfy the requirements in Table 2 after all vibration tests.

**Table 5 – Vibration condition**

| Range of vibration<br>Hz | Period<br>min | Vibration acceleration<br>$\text{m/s}^2$ | Amplitude<br>mm | Time<br>h  |            |            |
|--------------------------|---------------|------------------------------------------|-----------------|------------|------------|------------|
|                          |               |                                          |                 | Top-bottom | Left-right | Front-back |
| 5 ~ 200                  | 10            | 44,1                                     | <sup>a</sup>    | 4          | 2          | 2          |

<sup>a</sup> Amplitudes exceeding 10 mm shall be limited to 10 mm and amplitude shall be constant.

#### 7.5.6 Mechanical shock test

After testing the EVDR under the mechanical shock test presented in Table 6, the EVDR shall satisfy the requirements in Table 2. Note that the test specification below refers to IEC 60068-2-27.

**Table 6 – Shock condition**

| Waveform type | Peak acceleration<br>$\text{m/s}^2$ | Pulse duration<br>ms | Shock direction | Number of shocks |
|---------------|-------------------------------------|----------------------|-----------------|------------------|
| Half sine     | 490                                 | 11                   | X, Y, Z         | $\pm 3$          |

## Annex A (informative)

### Test setup for horizontal and vertical FOV

The test setup for horizontal and vertical FOV is illustrated in Figure A.1 and Figure A.2. Since the central point of video acquisition in the vertical direction may vary, the camera is fixed to allow a variation of  $\pm 10^\circ$  of the elevation angle. A square is placed on a fixed plate which is positioned 1 m from the camera excluding the lens. The size of the square is 1,678 m by 0,932 m. The EVDR passes the FOV test if all four corners of the square are shown in the video data. Note that when the distance between the EVDR camera and the plate is changed for test convenience, the size of the square can be varied with the same ratio.

Dimensions in metres

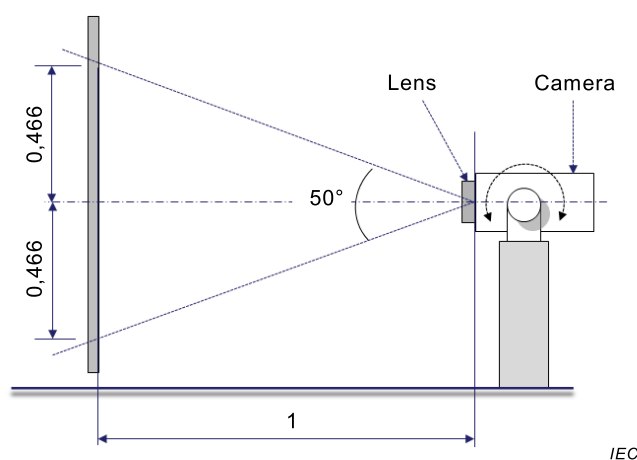


Figure A.1 – Test setup for horizontal and vertical FOV (front view)

Dimensions in metres

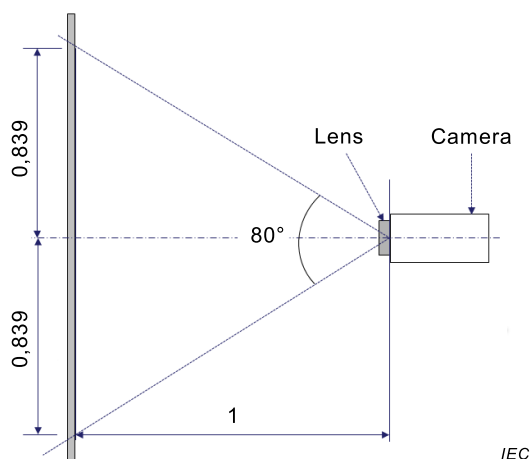


Figure A.2 – Test setup for horizontal and vertical FOV (plan view)

## **Annex B**

(informative)

### **Assignment of product identification number**

The 44-bit product identification number consists of the elements given in Table B.1.

**Table B.1 – Assignment of product identification number**

| <b>Bit</b>                                                                                                         | <b>Assignment</b>                  | <b>Number of combinations</b> | <b>Description</b>                                                                            |
|--------------------------------------------------------------------------------------------------------------------|------------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------|
| 1                                                                                                                  | 0                                  | Fixed                         | Use '0'                                                                                       |
| 2 to 9                                                                                                             | Manufacturer identification number | 256                           | A number assigned by IEC                                                                      |
| 10 to 15                                                                                                           | 1                                  | 64                            | Use '1'                                                                                       |
| 16 to 44                                                                                                           | Number assigned by manufacturer    | 536 870 912                   | A product number assigned by the manufacturer; the same number cannot be reused or duplicated |
| The product identification number displayed on the system can be expressed in binary, decimal or hexadecimal form. |                                    |                               |                                                                                               |

## Annex C (normative)

### Registration plate identification of the camera

The registration plate identification equipment can be set up as shown in Figure C.1. Since the central point of video acquisition in the vertical direction may vary, the camera is fixed to allow a variation of  $\pm 10^\circ$  of the pitch angle. A fixed plate is positioned 2 m from the camera, excluding the lens. A 4× magnified ISO 12233 resolution chart (see Figure C.2) is placed in the middle of the plate. The user converts the stored video to still images, and checks that the resolution is  $0,72 \pm 0,1$  through spatial frequency response (SFR) analysis.

NOTE The thickness of characters of a license plate 5 m away is  $(7,2 \pm 1)$  mm. A resolution of 0,72 is used because of the 4× magnified chart and the position of the plate at 2 m from the camera.

Dimensions in metres

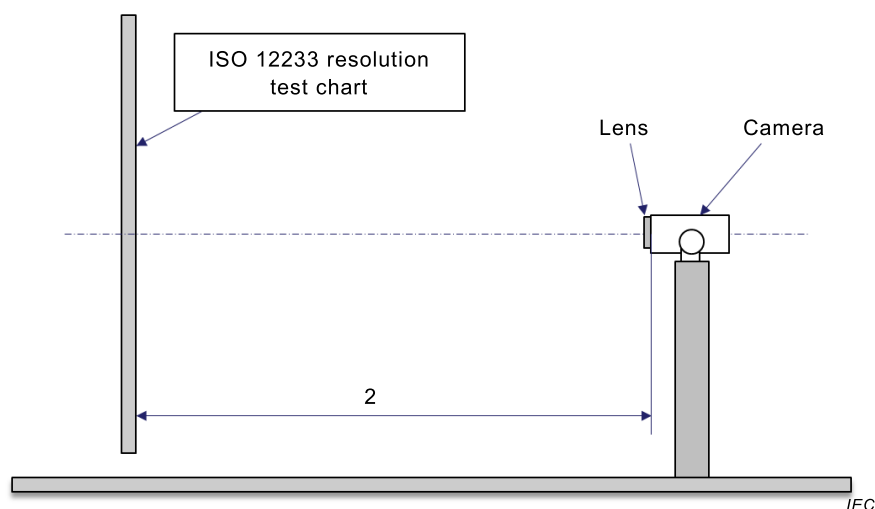


Figure C.1 – License plate identification test using the ISO 12233 resolution chart

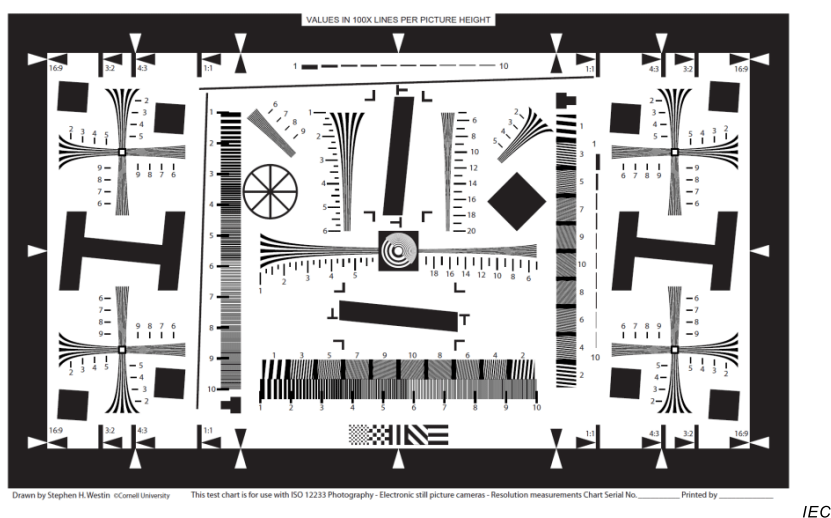


Figure C.2 – ISO 12233 resolution chart

## Bibliography

IEC 62133, *Secondary cells and batteries containing alkaline or other non-acid electrolytes – Safety requirements for portable sealed secondary cells, and for batteries made from them, for use in portable applications*

ISO 16844-7, *Road vehicles – Tachograph systems – Part 7: Parameters*

IEEE 1616, *IEEE Standard for Motor Vehicle Event Data Recorder (MVEDR)*

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